# Solution Architecture

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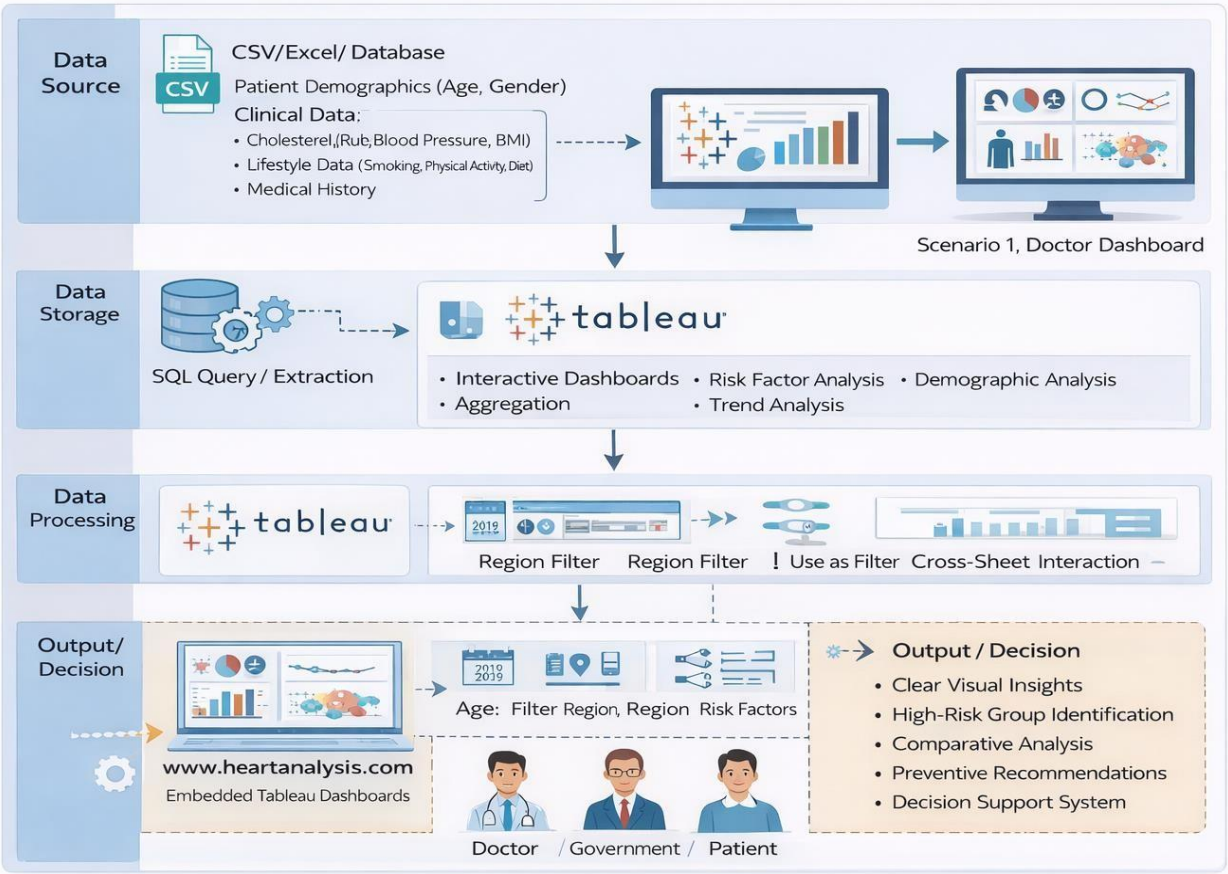
| Date | 5 January 2026 |
| Team ID | LTVIP2026TMIDS47882 |
| Project Name | Heart Disease Analysis |
| Maximum Marks | 4 Marks |

## Solution Architecture:

The Solution Architecture describes the overall structural design of the proposed Heart Disease Analysis system. It explains how healthcare data is collected, stored, processed, and transformed into meaningful visual insights using Tableau.

The architecture defines the complete flow of data from the data source layer to the visualization and dashboard layer, ensuring accuracy, scalability, and efficient decisionmaking. This structured design enables doctors, policymakers, and individuals to analyze heart disease risk factors effectively and take preventive actions.

## Example - Solution Architecture Diagram:


Data Source
CSV
CSV/Excel/ Database
Patient Demographics (Age, Gender)
Clinical Data:
• Cholesterel,(Rub,Blood Pressure, BMI)
• Lifestyle Data (Smoking, Physical Activty, Diet)
• Medical History
Scenario 1, Doctor Dashboard
Data Storage
SQL Query / Extraction
tableau
• Interactive Dashboards
• Risk Factor Analysis
• Demographic Analysis
• Aggregation
• Trend Analysis
Data Processing
tableau
2019
Region Filter
Region Filter
! Use as Filter
Cross-Sheet Interaction
Output/ Decision
www.heartanalysis.com
Embedded Tableau Dashboards
2019
2019
Age: Filter Region, Region Risk Factors
Doctor / Government / Patient
Output / Decision
• Clear Visual Insights
• High-Risk Group Identification
• Comparative Analysis
• Preventive Recommendations
• Decision Support System